

Separating effect size from statistical evidence in network-proximity rankings under target-count asymmetry: a controlled liver-interactome audit

Antony Bevan¹, Sarputheen M², and Vijayakumar Thangavel Mahalingam^{2,*}

¹Department of Pharmacognosy, Madurai Medical College, Madurai, India

²Department of Pharmacy Practice, SRM College of Pharmacy, Faculty of Medicine and Health Science, SRM Institute of Science and Technology, Kattankulathur, Chennai, India

*Corresponding author: Vijayakumar Thangavel Mahalingam; Email: vijayakm2@srmist.edu.in

Abstract

In network medicine, the proximity Z-score calibrates the interactome distance between a compound’s targets and a disease module against a degree-matched null, and its magnitude is compared across compounds to prioritize candidates. We show that such comparisons can obscure two separable quantities: standardized evidence and topological effect size. The null standard deviation of averaged target–disease statistics follows a scaling law, $\sigma_{\text{null}} \propto |T|^{-1/2}$, across shortest-path proximity, random-walk influence, and expression-weighted influence. In a liver-expressed STRING interactome, Hyperforin (10 targets) lies closer to the drug-induced liver injury (DILI) module than Quercetin (62 targets) ($d_c = 1.30$ vs 1.68 , $\text{STRING} \geq 900$) yet has the weaker proximity Z-score (-3.86 vs -5.44), and the Z-score ordering, but not the raw-distance ordering, reverses between STRING thresholds. A degree-controlled benchmark using 20,000 probes per target-set size locates material reversals within a narrow, high-margin operating regime rather than as a generic outcome. As a complementary effect-size readout, we use perturbation efficiency — the mean per-target random-walk influence — and decompose the Hyperforin signal into direct target–DILI overlap (62% of its efficiency) and a modest propagated advantage over Quercetin ($\sim 1.5\times$). Prioritizing compounds with unequal target counts therefore requires preserving the distinction between raw effect size, standardized evidence, and direct versus propagated influence.

Keywords: network medicine; network proximity; proximity Z-score; target-count asymmetry; effect size; random walk with restart; drug-induced liver injury.

1 Introduction

Network-based drug prioritization maps compound targets onto the human interactome and scores their proximity to disease-associated protein modules [1–4]. The field standard, the proximity Z-score, calibrates each compound against a size- and degree-matched null and answers the question: is this compound unusually close to this disease? Guney et al. [4] validated this Z-score as a per-pair classifier across 402 known drug–disease associations (mean 3.5 targets per drug) and showed that, within each drug’s own null, the standardization removes the dependence that raw distance has on target count and degree.

Comparing Z-score *magnitudes* across compounds, however, introduces a distinct requirement: the Z-score must be monotone in the underlying topological effect. When target counts differ sharply, the larger set’s null standard deviation shrinks as $|T|^{-1/2}$, a consequence of the law of large numbers on an averaged statistic, so a compound with weaker raw proximity can register stronger standardized evidence. The American Statistical Association emphasizes that statistical significance “does not measure the size of an effect or the importance of a result” [5]; whether this non-monotonicity materializes in network-medicine practice, and under what conditions, has not been systematically examined.

We address this with a controlled stress test in a liver-expressed interactome, using two *Hypericum perforatum* constituents as a deliberately high-contrast diagnostic pair: Hyperforin (10 targets) and Quercetin (62 targets). Hyperforin is the PXR-activating, CYP-inducing constituent responsible for St John’s Wort’s clinically important hepatic drug–drug interactions [6–8], which makes it a mechanistically motivated comparator for engagement of the DILI module’s xenobiotic-metabolism axis; Quercetin provides the high-target-count comparator. The drug-induced liver injury (DILI) module serves as the disease reference. Because it includes cytochrome-P450, transporter, and nuclear-receptor genes central to xenobiotic metabolism, transport, and regulatory response, it also exposes a second question: when does direct target–disease overlap inflate apparent network influence? We complement the pair with a degree-controlled calibration benchmark (20,000 probes per size, 500,000 cross-size pairs) that characterizes the operating regime in which rank reversal can occur, and we introduce perturbation efficiency, the mean per-target random-walk influence on the disease module, as a complementary effect-size measure. Target provenance, botanical context, and chemical-similarity controls are detailed in Methods and §2.7.

Together these show that proximity Z-score magnitude and raw closest distance can dissociate under target-count asymmetry; we trace this to the $|T|^{-1/2}$ null-precision law and find that the law holds for shortest-path, random-walk, and expression-weighted influence alike. A degree-controlled liver-network benchmark then locates material rank reversal as a conditional outcome: rare for generic target sets, and realized when the smaller set ranks in the top decile of probe-pair margins. On the diagnostic pair, perturbation efficiency ranks the PXR/CYP-inducing constituent higher, while the direct–propagated decomposition separates the direct target–DILI overlap from the propagated influence with which a headline number would conflate it. These two threads are one caution: a single network-proximity number can mislead a cross-compound comparison in two distinct ways, through size-dependent null precision and through distance-zero target–disease overlap. This is a controlled audit of how size-dependent null precision affects the interpretation of network-proximity rankings, not a DILI-prediction model.

2 Results

2.1 The null standard deviation shrinks with target count for every metric

The null standard deviation of the proximity Z-score follows a general property of averaged statistics: it shrinks as the number of targets grows. Drawing uniformly random seed sets of increasing size and measuring the null standard deviation of per-target influence, we find

$\sigma_{\text{null}} \propto |T|^{-0.47}$ (log-log slope; theoretical $-1/2$; Table 1); the degree-controlled benchmark of §2.2 refines this exponent to -0.499 (95% CI $[-0.502, -0.495]$). This is the Law of Large Numbers acting on a mean over targets, and it is observed across all three evaluated metrics: in the compound-specific degree-matched permutation nulls, the Hyperforin-to-Quercetin null-SD ratio is 2.40–2.57 for shortest-path, 2.45–3.04 for random-walk influence, and 2.47–2.93 for expression-weighted influence, against the expected $\sqrt{62/10} = 2.49$ (per-metric null parameters in Table S3). For shortest-path the ratio brackets $\sqrt{62/10}$ within finite- n estimation error ($n = 1,000$); for random-walk and expression-weighted influence it runs somewhat higher (up to 3.04), consistent with the heavier-tailed, degree-sensitive nulls of diffusion-based statistics [9, 10], whose standard deviations are estimated with larger finite- n error. The $|T|^{-1/2}$ scaling itself is confirmed independently by the degree-controlled benchmark (-0.499), not by these per-metric ratios. Influence-based Z-scores are therefore subject to the same shrinkage as proximity Z-scores, and Z-score magnitude should not be interpreted as a direct effect-size comparison across compounds of very different target count; this is why we pair the evidence with an explicit effect size. We first characterize the operating regime in which the null-precision law produces rank reversal.

Table 1. Null standard deviation of per-target influence scales as $|T|^{-1/2}$ (STRING ≥ 900 ; uniformly random seed sets). The log-log slope is -0.47 .

$ T $	5	10	20	40	62
σ_{null}	0.0093	0.0068	0.0059	0.0035	0.0028

2.2 The null-precision law defines a conditional operating regime for rank reversal

To characterize the operating regime of the proximity Z-score under target-count asymmetry, we constructed a degree-controlled calibration benchmark on the same liver LCC and DILI module, sampling random target-set “probes” across sizes ($|T| \in \{5, 8, 10, 15, 20, 30, 40, 60, 62, 80\}$) and comparing their raw closest distance with a size-level calibration Z-score; no toxicity outcome is modeled (Methods; `scripts/run_operating_regime_benchmark.py`). This benchmark Z-score is used to study the operating behavior of standardization across target counts; it is not the per-compound Guney null used for the real compounds.

The null-precision mechanism is also observed in size-matched pseudo-module controls. The null standard deviation scales as $\sigma_{\text{null}} \propto |T|^{-0.499}$ (95% CI $[-0.502, -0.495]$, $R^2 = 0.9999$; degree-distribution-pinned probes on a ten-point size grid, consistent with the coarser per-target-influence estimate of Table 1), indistinguishable from the theoretical $-1/2$ and from a uniform-probe control (-0.500), and the exponent is similar for the real DILI module (-0.495 ; Fig. 3A) and for random size-matched pseudo-modules (-0.498 ; Table S4). The larger set’s sharper null therefore gives it the *capacity* to register weaker absolute proximity as stronger standardized evidence: the maximum raw-distance margin it can overturn is $\delta_{\text{max}}(R) = (\mu_L - \mu_S) + |z_S| \sigma_L (\sqrt{R} - 1)$ (Fig. 3B; this follows by solving $Z_L = Z_S$ for the raw margin $\delta = d_c^L - d_c^S$, using $\sigma_S/\sigma_L = \sqrt{R}$ from the $|T|^{-1/2}$ law). The variance contrast scales as $\sqrt{R}$; for a fixed smaller-set size, the over-

turnable raw-margin term can be written as $|z_S|\sigma_S(1 - R^{-1/2})$, so it increases with target-count ratio but saturates. The coefficients are empirical: here the null-mean offset $\mu_L - \mu_S$ is negligible (+0.001), a property of this interactome’s distance geometry rather than a structural zero.

Realized reversals are conditional. Using descriptive material-margin cutoffs for the smaller set’s raw-distance advantage (≥ 0.3 hops, approximately $0.3/2.2 \approx 14\%$ of the null-mean closest distance in this network; ≥ 0.5 hops, $\sim 23\%$, as a stricter alternative), the larger set overturns the ranking in $\approx 0\%$ of degree-matched probe pairs up to $R = 4$ and only 0.39% at $R = 8$ (≥ 0.5 hops: $\leq 0.01\%$ throughout; a no-shrinkage counterfactual gives 0% ; Table S4). Unconditionally, however, Z-score and raw-distance rankings disagree in $\sim 12\%$ of cross-size comparisons at $R = 6.2$, a rate that is non-negligible for naive cross-compound Z-score comparison. In this benchmark, material reversals are concentrated when the smaller set is above the 90th percentile of probe-pair margins (large $|z_S|$), so that a modest raw advantage falls under the $\delta_{\max}$ envelope. A biologically interpretable pair falling within this regime is examined next (§2.3).

2.3 Raw proximity and proximity-Z rank the two compounds differently

The Hyperforin/Quercetin pair falls within the characterized regime. Hyperforin’s closeness margin over Quercetin (d_c difference 0.38) sits at the 91st percentile of probe-pair margins and below the $\delta_{\max} \approx 0.63$ envelope at $R = 6.2$, placing it inside the reversal region (Fig. 3C). In the liver-expressed largest connected component (LCC) of the STRING ≥ 900 interactome (7,677 nodes; 66,908 edges), Hyperforin engages 10 targets and Quercetin 62; the drug-induced liver injury (DILI) module contains 82 genes (Fig. 1; the 82-gene set is listed in Table S16). Hyperforin’s targets are topologically closer to the DILI module by shortest-path distance than Quercetin’s, $d_c = 1.30$ versus 1.68. The degree-matched proximity Z-scores order the compounds in the opposite direction: Hyperforin $Z = -3.86$, Quercetin $Z = -5.44$ (Table 2).

The Z-score is a standardized quantity, $Z = (d_c - \mu_{\text{null}})/\sigma_{\text{null}}$, in which the observed distance is compared to the mean and spread of distances for random target sets of the same size and degree. These two statistics answer different questions: d_c measures raw topological effect; Z measures standardized evidence against a null. Quercetin’s 62 targets sit far below the tightly concentrated null expected for 62 random degree-matched genes ($\mu = 2.17$, $\sigma = 0.091$), producing a large $|Z|$; Hyperforin’s 10 targets sit below a broader null ($\mu = 2.21$, $\sigma = 0.235$), producing a smaller $|Z|$ despite a shorter absolute distance. Hyperforin has the stronger raw closest-distance (d_c) effect; Quercetin has the stronger standardized evidence (Fig. 2). The instability is in the cross-compound *evidence ranking*: between thresholds the proximity-Z ranking reverses (Hyperforin $Z = -6.04$ vs Quercetin -5.46 at ≥ 700 ; the order flips at ≥ 900), whereas the raw d_c ranking (Hyperforin closer at both thresholds) does not.

Table 2. Effect size and standardized evidence dissociate under target-count asymmetry (STRING ≥ 900). Hyperforin is closer (d_c) yet less “significant” (Z); the difference is driven by the smaller null standard deviation of the larger target set.

Compound	n targets	d_c	μ_{null}	σ_{null}	Z
Hyperforin	10	1.30	2.21	0.235	-3.86
Quercetin	62	1.68	2.17	0.091	-5.44

2.4 Perturbation efficiency is the mean per-target influence

We quantify influence with random walk with restart (RWR) [9, 11]. With column-normalized transition matrix $W = AD^{-1}$, restart probability α , and restart vector $\mathbf{p}_0$, let $\mathbf{I}_n$ denote the $n \times n$ identity matrix. The steady state is

$$\mathbf{p} = \alpha [\mathbf{I}_n - (1 - \alpha)W]^{-1} \mathbf{p}_0. \quad (1)$$

Influence on the DILI module D is

$$I(T, D) = \sum_{d \in D} p_d. \quad (2)$$

Taking the restart vector uniform over the target set, $p_0(i) = 1/|T|$ for $i \in T$ and 0 otherwise, Eq. (1) is linear in $\mathbf{p}_0$, so

$$E(T, D) \equiv I(T, D) = \frac{1}{|T|} \sum_{t \in T} \sum_{d \in D} p_d^{(t)}, \quad (3)$$

Perturbation efficiency E is therefore exactly the mean per-target influence on the disease module — a size-normalized effect-size measure, not a re-standardized evidence statistic. Here $\mathbf{p}^{(t)}$ denotes the RWR steady-state vector seeded from target t alone. We verified Eq. (3) numerically (joint vs mean-of-single-target influence agree to RWR tolerance: 0.11380975 vs 0.11380968 for Hyperforin).

At $\text{STRING} \geq 900$, $E = 0.1138$ for Hyperforin and 0.0322 for Quercetin (Fig. 4; and 0.1212 vs 0.0326 at ≥ 700): unlike the proximity-Z ranking, which reverses between thresholds (§2.3), the per-target influence ranking is stable across the two evaluated network thresholds for both RWR and expression-weighted influence (Table S6). The ranking is robust to the restart probability across $\alpha \in [0.10, 0.70]$ (Hyperforin above Quercetin throughout), although the magnitude of the ratio is α -dependent (2.90 at $\alpha = 0.10$ to 13.35 at $\alpha = 0.70$; Table S7); we therefore treat the ranking, not the ratio, as the robust feature, and do not interpret α -sensitive fold ratios as absolute biological constants. Expression weighting (destination-node liver expression; Methods) preserves the ordering (Fig. 5) and is insensitive to the expression floor (efficiency ratio 2.69–2.70 across floors 0–0.1; Table S8).

2.5 Separating direct overlap from propagated influence

The raw per-target advantage ($3.5\times$) conflates two mechanistically distinct contributions. We isolate the propagated component and then stress-test it through a chain of controls: an orthogonal separation measure, per-target distance-1 connectivity, removal of the overlapping module genes, and a size-matched bootstrap. Four of ten Hyperforin targets (ABCB1, CYP2C9, MMP2, NR1I2) are themselves DILI-module genes (Tables S1, S2), versus one of sixty-two for Quercetin (MMP2), so part of Hyperforin’s measured disease-module influence is *direct*: steady-state probability assigned to seed nodes that are themselves DILI-module genes. The canonical proximity framework retains such distance-zero contributions, treating a drug target that coincides with a disease protein as a real, if special, signal [4]; we retain it likewise, but separate it from indirect,

network-*propagated* influence. Analogous to leave-one-out scoring used in network propagation [9, 11], we score each target against the disease module *excluding the target itself*, isolating the propagated part of the module influence (Eq. (2)).

Decomposed this way (Table 3, Fig. 6), 62% of Hyperforin’s per-target influence is direct overlap (direct component 0.0711 versus 0.0032 for Quercetin, only 10% of which is direct), while its propagated component is 0.0427 versus 0.0290, a propagated advantage of $1.5\times$ that is far below the raw $3.5\times$. The propagated component remains above a degree-matched background: it sits at the 99.9th percentile of a degree-matched random 10-gene background ($n = 1,000$; $3.3\times$ the background mean, empirical $p = 0.002$), although it does not exceed every random set, and some size-matched Quercetin subsets reach comparable values; alternative exclusions of the propagated term bound the advantage to $1.2\text{--}1.9\times$. Both components concentrate in the PXR–CYP–transporter axis, which is at once the established mechanism of Hyperforin-mediated hepatic interactions and the locus of the overlap. The propagated advantage is therefore modest, layered on a larger and mechanistically coherent direct-overlap effect rather than reflecting a clear topological separation. Two further topology checks clarify this interpretation: the network separation measure S_{AB} [3] shows both target sets are topologically separated from the DILI module, Hyperforin slightly more so (Table S10). This does not contradict the d_c result because S_{AB} also depends on within-set compactness; Hyperforin’s target set is tighter internally ($\langle d_{AA} \rangle = 1.20$ vs 1.55), raising its separation despite its shorter closest distance. Separately, Hyperforin’s targets carry more direct (distance-1) DILI connections per target (3.4 vs 1.5 ; Table S11), so the advantage is concentrated in a few directly DILI-linked targets. The decomposition is also robust to the disease-module definition itself: removing the four entangled pharmacogenes from the module, so the targets cannot directly overlap it, leaves the propagated advantage intact and marginally larger ($1.47\times \rightarrow 1.58\times$; Table S12), showing the propagated residual is not eliminated when direct overlap is removed from the module. A size-matched bootstrap of the raw influence (Fig. S1) reproduces the baseline ratio but, lacking the overlap adjustment, is superseded by this decomposition.

Table 3. Direct versus propagated decomposition of per-target influence (STRING ≥ 900). Raw influence separates into a direct component (steady-state mass on $\text{target} \cap \text{DILI}$ nodes, retained following Guney et al. [4]) and a propagated component isolated by leave-one-out [11]; alternative exclusions of the propagated term bound the residual advantage. The direct-overlap ratio is reported descriptively (computed from full-precision values, so it need not equal the quotient of the rounded entries shown) and is not interpreted as a biological fold-effect.

Component / control	$E(\text{Hyperforin})$	$E(\text{Quercetin})$	ratio
Raw per-target influence	0.1138	0.0322	3.5
Mass on $\text{target} \cap \text{DILI}$ nodes	0.0711	0.0032	22.5
Propagated (leave-one-out)	0.0427	0.0290	1.5
Propagated, CYP/transporter targets removed	0.0338	0.0290	1.2
Propagated, shared targets removed	0.0541	0.0288	1.9

2.6 The proximity computation is faithful to the canonical implementation

We next compared the proximity result against Guney et al.’s canonical toolbox implementation: the closest distance reproduces exactly (observed d_c 1.300/1.677) and the Z-scores to within permutation tolerance. All reference distributions use $n = 1,000$ random sets, fixed seed 42, Guney’s ≥ 100 -node degree-binning, and preserve the DILI module size ($|D| = 82$). Under the fixed-disease null (randomizing targets only, disease module held at the 82 DILI genes), the ≥ 100 -node binning gives Hyperforin $Z = -4.09$, Quercetin $Z = -5.34$ — Quercetin more “significant” despite being topologically farther by shortest-path distance. Under Guney’s two-sided null (randomizing both the target set and a degree-matched disease set of size 82), both compounds remain strongly proximal relative to their nulls but the dissociation attenuates to near-parity: Hyperforin $Z = -3.55$, Quercetin $Z = -3.66$ (the null standard deviation continues to shrink with set size, ratio 2.37 vs expected 2.49). The interpretation is consistent: Hyperforin is the topologically closer compound in every construction, while its evidence ranking relative to Quercetin depends on the null — strongest when the actual DILI module is held fixed, the configuration relevant to comparing two compounds against one disease (Table 4).

Table 4. Revalidation against the canonical proximity implementation (STRING ≥ 900 , $n = 1,000$, seed 42, ≥ 100 -node degree-binning, $|D| = 82$ preserved). The effect/evidence dissociation is clear under the fixed-disease null and attenuates under the two-sided null; Hyperforin is the closer compound throughout. The raw d_c ordering (Hyperforin closer) is invariant across all constructions; the Z-score evidence difference is null-dependent.

Null model (d_c)	Hyperforin Z	Quercetin Z
Repo $\pm 25\%$ window, fixed disease (primary)	-3.86	-5.44
Guney ≥ 100 -bin, fixed disease	-4.09	-5.34
Guney ≥ 100 -bin, two-sided	-3.55	-3.66

2.7 Chemical-similarity specificity control

Neither compound is a close structural analogue of DILIrank 2.0 reference drugs [12, 13]—the retrievable-SMILES binary subset of DILIrank 2.0 (542 DILI-positive, vMost/vLess-concern, and 365 DILI-negative, vNo-concern, drugs; the 354 ambiguous-concern drugs are excluded; Methods)—with maximum Tanimoto to any DILIrank reference drug < 0.4 (Hyperforin 0.20, Quercetin 0.22; the maxima against DILI-positive references alone are 0.15 and 0.21 [14]), so structural-analogue similarity to those compounds is unlikely to explain the observed network pattern (Fig. 7); this argues against close structural-analogue confounding within the DILIrank reference set.

3 Discussion

Our results build on the proximity framework established by Guney et al. [4]: the proximity Z-score is a calibrated *evidence* statistic in which each compound is compared to a size- and degree-matched null, and a larger target set legitimately yields sharper significance. The phenomenon we characterize arises when the task shifts from per-compound classification to

cross-compound ranking. Under target-count asymmetry, Z-score *magnitude* should not be read as a biological effect-size ordering. Because the Z-score’s denominator shrinks as $|T|^{-1/2}$ (a property we observe for shortest-path, random-walk, and expression-weighted influence alike), the evidence ranking can diverge from, and even reverse, the raw topological effect ranking. This is the established distinction between effect size and statistical significance emphasized by the American Statistical Association [5], here in a network-medicine setting. The magnitude of the dissociation is itself null-dependent: it is largest under the fixed-disease null appropriate to comparing two compounds against one disease module, and attenuates to near-parity under Guney’s population-level two-sided null (Hyperforin $Z = -3.55$, Quercetin $Z = -3.66$; Table 4).

The contribution here is not that advice but a quantitative map of when it bites: the $|T|^{-1/2}$ null-precision characterization across three influence metrics, the analytic overturn envelope $\delta_{\max}(R)$ and its saturation, a degree-controlled benchmark that quantifies how often material reversal occurs, and a decomposition that separates distance-zero overlap from propagated positioning. The practical consequence is to report an explicit effect size alongside the evidence Z-score. We use per-target random-walk influence (perturbation efficiency), which by linearity of the walk equals the mean single-target influence on the disease module and is therefore comparable across compounds with unequal target counts. This metric complements the Z-score rather than replacing it: it does not itself escape variance shrinkage at the level of its own Z-score. Random-walk influence is also sensitive to node degree [9, 10], which is one reason we decompose it into direct-overlap and propagated components below.

Perturbation efficiency ranks Hyperforin above Quercetin. The relevant reference ordering here is engagement of the xenobiotic-metabolism axis, not hepatotoxic potential: Hyperforin is the PXR-activating, CYP-inducing constituent behind St John’s Wort’s hepatic drug–drug interactions [6–8], and the claim is that E recovers that mechanistic ordering rather than ranking DILI risk. This ordering holds across restart-probability variation, network threshold, and degree-binning. Decomposing the advantage, however, shows that it is largely *direct* target–disease overlap: 62% of Hyperforin’s per-target influence comes from four targets that are themselves DILI-module genes. The *propagated* component, isolated by leave-one-out, is a modest $1.5\times$ (about $1.2\text{--}1.9\times$ across alternative exclusions); its distribution overlaps size-matched Quercetin subsets, though it remains above a degree-matched background. Both components concentrate in the xenobiotic-metabolism axis that defines the mechanism and the overlap alike. Hyperforin’s propagated advantage is above a degree-matched background (99.9th percentile, $3.3\times$ the mean) and above the mean of size-matched Quercetin subsets, while still overlapping their upper tail; the direct-overlap component is mechanistically interpretable but not independent evidence of propagated network positioning.

3.1 Relationship to prior work

Our findings build on Guney et al.’s proximity framework [4] by identifying a cross-compound ranking regime that the original per-pair study design did not stress-test. Guney et al. evaluated proximity as a per-pair classifier across 238 drugs and 78 diseases (402 known drug–disease associations; mean 3.5 targets per drug), and demonstrated that the closest-distance relative proximity is essentially uncorrelated with target degree ($\rho = -0.01$); that is a statement about

classification calibration within each drug’s own size-matched null, not about the comparability of *Z-magnitudes* across compounds whose target counts differ several-fold. Our task is different: cross-compound ranking under extreme target-count asymmetry (10 vs 62). In that regime the size-dependence of the null standard deviation, which the per-pair Z-score calibrates through its null mean and null variance, re-emerges when Z-magnitudes are compared between compounds. The resolution is not a new significance statistic but the separate reporting of effect size, for which per-target influence is a natural choice. Random walk with restart is conceptually adjacent to attenuated-distance and diffusion-style measures, but it is not identical to Guney’s diffusion-kernel proximity (d_k), which downweights longer *shortest paths* by an exponential penalty rather than propagating restart-anchored probability mass over the adjacency matrix; we therefore do not use Guney’s d_k result to argue for or against random-walk influence. The choice between proximity measures is orthogonal to our argument: a normalized influence measure can define an effect-size scale that should be reported *alongside* the standardized evidence rather than in place of it when compounds of unequal size are compared. Degree bias in network propagation and prioritization is well documented [9, 10, 15]; the operating-regime benchmark (§2.2) complements node-degree control by explicitly calibrating the target-count axis, establishing that the $|T|^{-1/2}$ null-precision law is stable across the DILI module and size-matched pseudo-modules and that the resulting effect/evidence reversal is a *located, conditional* regime, materializing mainly for smaller sets in the top decile of probe-pair margins.

3.2 Limitations

This study is a controlled two-compound audit designed to characterize the statistical behavior of proximity Z-scores under target-count asymmetry; it does not evaluate toxicological outcomes or predict DILI. Hyperforin is not an established intrinsic hepatotoxin [16]; the pair was selected for target-count contrast and mechanistic interpretability. Network influence is a measure of topological reach, not a toxicological outcome: it omits dose, exposure, pharmacokinetics, binding directionality (agonism vs antagonism), and reactivity. The analysis should therefore be read as an estimand audit, not a compound-risk classifier. The two target sets also differ in provenance, not only size: Hyperforin’s ten targets are literature-curated and mechanistically central (the PXR receptor with its induced enzymes and transporters), whereas Quercetin’s sixty-two are a broad ChEMBL bioactivity net (Table S1). This evidentiary asymmetry could in principle contribute to Hyperforin’s apparent advantage independently of topology; the direct-propagated decomposition and the DILI-module sensitivity (Table S12) are designed to bound that confound (the propagated advantage survives both), but they do not eliminate it, and a fully curation-matched comparison would require harmonized target evidence for both compounds, which we note as a limitation. Accordingly, target-provenance asymmetry limits the biological interpretation of the worked pair more than the statistical calibration, which is estimated from degree-controlled probes rather than from the curated compounds. The ChEMBL-derived Quercetin target set may also include assay-context targets that differ from clinically relevant exposure targets. Generalization to compound libraries, for example testing whether perturbation efficiency carries DILI signal beyond target count and degree across DILIRank 2.0 with curated target sets, is a direction for future work. The operating-regime benchmark (§2.2) establishes

that the $|T|^{-1/2}$ exponent holds for both the DILI module and matched pseudo-modules and that the reversal is conditional; the *rate* of reversal depends on the module’s distance geometry and would differ in other networks. Its probes are degree-controlled random sets rather than curated drug-target sets, so it characterizes the statistic, not a drug population. Finally, interactome incompleteness and degree bias in random-walk propagation [9, 10] bound the precision of all network-based statements made here.

3.3 Conclusion

Under target-count asymmetry, proximity Z-score magnitude is evidence strength, not an effect-size ranking. This distinction matters because a valid, sharper null can make a larger target set appear more strongly supported even when its raw proximity is weaker. Reporting raw effect size, standardized evidence, and per-target influence decomposed into direct-overlap and propagated components separately yields a more transparent comparative picture. A degree-controlled liver-network calibration shows that the null-precision law holds for the module and its size-matched controls, and that material rank reversal is a conditional regime that emerges for a high-margin (top-decile) smaller set. In the worked example, per-target influence ranks the PXR/CYP-inducing constituent higher, while the decomposition separates direct target–disease overlap from propagated influence.

4 Methods

4.1 Interactome and modules

Human protein functional associations were obtained from the STRING v12.0 functional association network [17] (combined score integrating experimental, curated-database, co-expression, genomic-context, and text-mining channels; we use “interactome” in this broad association-network sense, not as a purely physical binding network) at combined-confidence ≥ 900 , mapped to HUGO gene symbols, and filtered to genes with GTEx v8 liver median TPM ≥ 1 [18]; the largest connected component (LCC) was retained using NetworkX [19] (7,677 nodes, 66,908 edges, from a raw ≥ 900 STRING network of 11,693 nodes / 100,383 edges; the ≥ 700 network, 9,773 LCC nodes, is used for threshold robustness). DILI-associated genes were obtained from DisGeNET curated associations (UMLS C0860207) [20], giving 82 genes in the ≥ 900 LCC; this is a curated association module, not a validated causal gene set. Hyperforin targets (10 in the LCC) were curated from primary literature [6, 7, 21–27]; Quercetin targets (62 in the LCC) were assembled from ChEMBL-derived bioactivity records [28] (the programmatic retrieval is documented in the repository’s data manifest rather than re-executed here; see Data Availability). Protein identifiers were mapped to HUGO symbols via UniProt [29], with non-human proteins excluded and symbols standardized (e.g. MDR1→ABCB1); raw counts (14 Hyperforin, 122 Quercetin) reduce to 10 and 62 in the LCC (Table S15). Five genes are targeted by both compounds (ABCG2, AKT1, CYP3A4, MMP2, MMP9) and are retained in both sets; four further literature-reported Hyperforin targets (TRPC6, VEGFA, PMAIP1, GRIN1) fall outside the liver-expressed LCC and are not analyzed [30–32].

4.2 Closest distance and proximity Z-score

Closest distance is the mean shortest path from each target to the nearest disease-module gene [4]:

$$d_c(T, D) = \frac{1}{|T|} \sum_{t \in T} \min_{s \in D} \text{dist}(t, s). \quad (4)$$

The proximity Z-score standardizes this raw distance against a degree-matched null:

$$Z = \frac{d_c - \mu_{\text{null}}}{\sigma_{\text{null}}}, \quad (5)$$

with $Z = 0$ when $\sigma_{\text{null}} = 0$. We use $n = 1,000$ degree-matched random reference sets, fixed seed 42. Following Guney et al. [4] we report Z as the primary magnitude; we additionally report empirical, directional permutation p-values using the conservative $(r + 1)/(n + 1)$ convention [33] — one-sided for smaller-than-null distance (proximity) and larger-than-null influence (RWR/EWI) — whose floor at $n = 1,000$ is 9.99×10^{-4} (reported as $p < 0.001$). For both compounds, $p < 0.001$ confirms non-random proximity to the DILI module but does not discriminate between them; the Z-score is the primary comparative statistic. No two-tailed permutation p-value is used. Degree matching is performed both with a $\pm 25\%$ degree window (primary) and with Guney’s ≥ 100 -node degree-binning; the degree-binning choice has minimal effect under the fixed-disease null (Table S13). In the Guney-fidelity comparison, we use random seed 42 rather than the toolbox default (452456), and the local shortest-path comparison sampler permits the original seed node in its own candidate pool whereas the toolbox excludes it; this is a $\leq 1\%$ candidate-pool difference for bins of ≥ 100 nodes, and the resulting Z-scores are unchanged within permutation tolerance (Table S13). The shared RWR/EWI permutation sampler excludes original targets from the random candidate pool. The fixed-disease null (randomizing targets only) is primary, as it corresponds to comparing two compounds against one disease module; the two-sided null (randomizing both targets and disease, as in the canonical implementation) is reported as a robustness check.

4.3 Random walk with restart and perturbation efficiency

RWR uses $W = AD^{-1}$, restart $\alpha = 0.15$ (a PageRank-style damping value; the canonical RWR reference is Köhler et al. [11], where α is typically larger and results are reported robust to it; Guney et al. [4] use shortest-path proximity and define no restart parameter), convergence tolerance 10^{-6} , and ≤ 100 iterations. Influence and perturbation efficiency are defined by Eqs. (1)–(3); perturbation efficiency equals mean per-target influence by linearity of the steady state in the restart vector. Degree-matched permutation nulls ($n = 1,000$, seed 42) provide influence Z-scores and empirical p-values; Benjamini–Hochberg correction is applied within each network threshold. A full restart-probability sweep ($\alpha \in [0.10, 0.70]$) is reported.

4.4 Expression weighting

Transitions are weighted by destination-node liver expression, $W'_{ij} = A_{ij}e_i / \sum_k A_{kj}e_k$, with e_i the min–max-normalized $\log_2(\text{TPM} + 1)$ from GTEx v8 liver [18]; an expression floor of 0.01 maintains reachability and is shown to be immaterial (floor sweep 0–0.1).

4.5 Direct-propagated decomposition

To separate direct target-disease overlap from propagated influence, we isolate the propagated component by leave-one-out scoring: each target is scored against the module after excluding the target itself, thereby removing the disease-member seed’s own contribution from its module score. We then report a chain of controls: removal of DILI-member seeds, removal of cytochrome-P450/transporter seeds, removal of targets shared by both compounds, comparison with global degree-matched random 10-gene sets, and size-matched Quercetin subsets. The network separation measure S_{AB} [3] is reported as an orthogonal topology statistic. As a network-construction robustness check, we also rebuilt the interactome after excluding STRING edges supported solely by text mining [34] and re-ran all metrics, testing whether text-mining-only edges drive the observed effect ordering (Supplementary Note, *Text-mining robustness*).

4.6 Chemical similarity

Morgan fingerprints (ECFP4; radius 2, 2048 bits) were generated with RDKit [35]; maximum Tanimoto similarity to DILIrank 2.0 reference drugs [12, 13] was computed, with 0.4 as the structural-analogue threshold [14]. DILIrank 2.0 classifies 1,336 drugs into four concern categories; we used the binary subset with retrievable PubChem SMILES, mapping vMost- and vLess-DILI-concern to DILI-positive (542 drugs) and vNo-DILI-concern to DILI-negative (365 drugs), and excluding the ambiguous-DILI-concern category.

4.7 Operating-regime benchmark

To assess the statistical mechanism underlying the effect-size/evidence dissociation independently of the worked pair, we sampled random target-set probes on a ten-point size grid ($|T| \in \{5, 8, 10, 15, 20, 30, 40, 60, 62, 80\}$; 20,000 probes per size) from the liver LCC and computed each probe’s closest distance to the DILI module and its size-level calibration Z-score. Primary probes were degree-distribution-pinned to the full compound-target row degree profile (10 Hyperforin rows plus 62 Quercetin rows) using Guney’s ≥ 100 -node bins: for each probe, degree-bin labels were sampled with replacement from this target-row degree-bin distribution, and nodes were sampled from the corresponding bins after excluding DILI-module genes from the candidate pool so distance-zero mass could not scale with $|T|$. A non-DILI target-degree-profile family and a uniform-from-pool family were used as contrasts.

The null parameters (μ_{null} , σ_{null}) were estimated once per size and degree-profile family from the 20,000-probe reference distribution itself. Each probe’s calibration Z-score is therefore its deviation from that shared size-level reference, not a separate per-probe permutation null. Because $N = 20,000$, self-inclusion has negligible influence on size-level null parameters. We fitted the null-SD scaling exponent by log-log regression with bootstrap confidence intervals and compared it across the DILI module and three random size-matched pseudo-modules.

For the H/Q-matched operating-regime analysis, rank discordance was quantified on cross-size probe pairs with the smaller set fixed at $|T| = 10$ and the larger-set size varied (500,000 probe pairs per size-pair cell). We computed the maximum overturnable raw margin, $\delta_{\text{max}}(R) = (\mu_L - \mu_S) + |z_S| \sigma_L (\sqrt{R} - 1)$, and the margin-conditional reversal rate: the fraction of pairs

in which the smaller set is closer by at least δ_0 hops yet the larger set has the more negative calibration Z-score. The descriptive material-margin cutoffs were $\delta_0 \in \{0.3, 0.5\}$, with a no-shrinkage ($\sigma_L := \sigma_S$) counterfactual. These δ_0 values summarize material-margin behavior in the probe benchmark; they are not used to select or validate the worked pair, which is located separately by its observed margin, margin percentile, and $\delta_{\max}$ envelope.

Unconditional quantities are reported only as descriptive contrasts, with the H/Q-directional pattern (smaller set closer and larger set stronger by Z) distinguished from bidirectional rank discordance. This is a calibration benchmark of the statistic, not a drug-level toxicity predictor. Fixed seed 42; `scripts/run_operating_regime_benchmark.py`; outputs in `results/tables/operating_regime_*.csv`.

4.8 Reproducibility

All analyses run from the committed processed data with fixed seed 42, using Python 3.12 (NetworkX 3.6, NumPy 2.5, pandas 2.3, SciPy 1.18) and RDKit 2026.03; figures use R 4.6 with ggplot2 4.0. Exact pinned versions are in `requirements-lock.txt`. The committed pipeline and reviewer-evidence scripts regenerate the Results numbers from the deposited data.

Use of AI tools

AI-assisted tools were used for code drafting and language editing. All analyses, results, and interpretations were verified by the authors.

Data Availability

All processed data supporting this study (the committed liver-interactome networks, the curated target and DILI gene sets, and the computed result tables, with SHA-256 checksums in `data/CHECKSUMS.sha256`) are openly available in the project repository, and will be deposited as an archived Zenodo snapshot upon publication (see Code Availability): <https://github.com/antonbyevan/h-perforatum-network-tox-release>

Computed source data underlying the figures and tables are provided in the project repository; key tabulated summaries are included in the Supplementary Information. Raw data were obtained from the following public repositories:

- STRING v12.0: <https://string-db.org>
- GTEx v8: <https://gtexportal.org>
- ChEMBL v31: <https://www.ebi.ac.uk/chembl>
- DisGeNET: <https://www.disgenet.org>
- DILIrank 2.0: <https://www.fda.gov/science-research/ltk>

Code Availability

All analysis code — the network-toxicology pipeline, the perturbation-efficiency metric, the permutation, target–disease overlap (leakage) and Guney-fidelity analyses, and the figure scripts — is openly available under the MIT license at <https://github.com/antonybevan/h-perforatum-network-tox-release> (manuscript text, figures, and data are released under CC-BY-4.0). The version used for publication will be archived with a permanent Zenodo DOI from the tagged release before final submission. The deposited snapshot includes pinned dependency versions (`requirements-lock.txt`; Python 3.12), SHA-256 checksums for the committed processed inputs and result artefacts (`data/CHECKSUMS.sha256`), the full execution graph (`scripts/run_pipeline.py`), and verification scripts for reviewer-evidence, Guney-fidelity, leakage, and number-consistency checks.

References

- [1] Andrew L Hopkins. Network pharmacology: the next paradigm in drug discovery. *Nature Chemical Biology*, 4(11):682–690, 2008. doi: 10.1038/nchembio.118.
- [2] Albert-László Barabási, Natali Gulbahce, and Joseph Loscalzo. Network medicine: a network-based approach to human disease. *Nature Reviews Genetics*, 12(1):56–68, 2011. doi: 10.1038/nrg2918.
- [3] Jörg Menche, Amitabh Sharma, Maksim Kitsak, Susan Dina Ghiassian, Marc Vidal, Joseph Loscalzo, and Albert-László Barabási. Uncovering disease-disease relationships through the incomplete interactome. *Science*, 347(6224):1257601, 2015. doi: 10.1126/science.1257601.
- [4] Emre Guney, Jörg Menche, Marc Vidal, and Albert-László Barabási. Network-based in silico drug efficacy screening. *Nature Communications*, 7:10331, 2016. doi: 10.1038/ncomms10331.
- [5] Ronald L Wasserstein and Nicole A Lazar. The ASA statement on p-values: Context, process, and purpose. *The American Statistician*, 70(2):129–133, 2016. doi: 10.1080/00031305.2016.1154108.
- [6] Linda B Moore, Bryan Goodwin, Stacey A Jones, G Bruce Wisely, Connie J Serabjit-Singh, Timothy M Willson, John L Collins, and Steven A Kliewer. St. John’s wort induces hepatic drug metabolism through activation of the Pregnane X Receptor. *Proceedings of the National Academy of Sciences*, 97(13):7500–7502, 2000. doi: 10.1073/pnas.130155097.
- [7] Reginald E Watkins, G Bruce Wisely, Linda B Moore, John L Collins, Millard H Lambert, Shawn P Williams, Timothy M Willson, Steven A Kliewer, and Matthew R Redinbo. The human nuclear xenobiotic receptor PXR: structural determinants of directed promiscuity. *Science*, 292(5525):2329–2333, 2001. doi: 10.1126/science.1060762.
- [8] Siyu Chen, Xue Wang, Xinran Ye, Qinqin Wang, Xin Sun, Chunyan Ma, Zhidong Yuan, and Yang Yu. St. John’s wort exacerbates acetaminophen-induced liver injury through

PXR and CYP-mediated bioactivation. *Toxicological Sciences*, 190(1):68–80, 2022. doi: 10.1093/toxsci/kfac098.

[9] Lenore Cowen, Trey Ideker, Benjamin J Raphael, and Roded Sharan. Network propagation: a universal amplifier of genetic associations. *Nature Reviews Genetics*, 18(9):551–562, 2017. doi: 10.1038/nrg.2017.38.

[10] Sinan Erten, Gürkan Bebek, Rob M Ewing, and Mehmet Koyutürk. DADA: Degree-aware algorithms for network-based disease gene prioritization. *BioData Mining*, 4:19, 2011. doi: 10.1186/1756-0381-4-19.

[11] Sebastian Köhler, Sebastian Bauer, Denise Horn, and Peter N Robinson. Walking the interactome for prioritization of candidate disease genes. *The American Journal of Human Genetics*, 82(4):949–958, 2008. doi: 10.1016/j.ajhg.2008.02.013.

[12] Minjun Chen, Ayako Suzuki, Shraddha Thakkar, Ke Yu, Chao Hu, and Weida Tong. DILI-rank: the largest reference drug list ranked by the risk for developing drug-induced liver injury in humans. *Drug Discovery Today*, 21(4):648–653, 2016. doi: 10.1016/j.drudis.2016.02.015.

[13] AyoOluwa O Olubamiwa, Yanyan Qu, Skylar Connor, Weida Tong, Dongying Li, and Minjun Chen. DILRank 2.0: an updated and expanded database for drug-induced liver injury risk based on FDA labeling and a literature review. *Drug Discovery Today*, 30(11):104485, 2025. doi: 10.1016/j.drudis.2025.104485.

[14] Gerald Maggiora, Martin Vogt, Dagmar Stumpfe, and Jürgen Bajorath. Molecular similarity in medicinal chemistry. *Journal of Medicinal Chemistry*, 57(8):3186–3204, 2014. doi: 10.1021/jm401411z.

[15] Gal Barel and Ralf Herwig. NetCore: a network propagation approach using node coreness. *Nucleic Acids Research*, 48(17):e98, 2020. doi: 10.1093/nar/gkaa639.

[16] National Institute of Diabetes and Digestive and Kidney Diseases. LiverTox: Clinical and research information on drug-induced liver injury [internet]. St. John’s wort. <https://www.ncbi.nlm.nih.gov/books/NBK548880/>, 2018.

[17] Damian Szklarczyk, Rebecca Kirsch, Mikaela Koutrouli, Katerina Nastou, Farrokh Mehryary, Radja Hachilif, Annika L Gable, Tao Fang, Nadezhda T Doncheva, Sampo Pyysalo, Peer Bork, Lars J Jensen, and Christian von Mering. The STRING database in 2023: protein–protein association networks and functional enrichment analyses for any sequenced genome of interest. *Nucleic Acids Research*, 51(D1):D483–D489, 2023. doi: 10.1093/nar/gkac1000.

[18] GTEx Consortium. The GTEx Consortium atlas of genetic regulatory effects across human tissues. *Science*, 369(6509):1318–1330, 2020. doi: 10.1126/science.aaz1776.

[19] Aric A Hagberg, Daniel A Schult, and Pieter J Swart. Exploring network structure, dynamics, and function using NetworkX. In Gaël Varoquaux, Travis Vaught, and Jarrod Millman,

editors, *Proceedings of the 7th Python in Science Conference (SciPy 2008)*, pages 11–15, 2008.

[20] Janet Piñero, Juan Manuel Ramírez-Anguita, Josep Saüch-Pitarch, Francesco Ronzano, Emilio Centeno, Ferran Sanz, and Laura I Furlong. The DisGeNET knowledge platform for disease genomics: 2019 update. *Nucleic Acids Research*, 48(D1):D845–D855, 2020. doi: 10.1093/nar/gkz1021.

[21] R Scott Obach. Inhibition of human cytochrome P450 enzymes by constituents of St. John’s Wort, an herbal preparation used in the treatment of depression. *Journal of Pharmacology and Experimental Therapeutics*, 294(1):88–95, 2000. doi: 10.1124/jpet.294.1.88.

[22] Bernard J Komoroski, Shuyan Zhang, Steven A Wrighton, Stephen C Strom, Raman Venkataramanan, and Erin G Schuetz. Induction and inhibition of cytochromes P450 by the St. John’s wort constituent hyperforin in human hepatocytes. *Drug Metabolism and Disposition*, 32(5):512–518, 2004. doi: 10.1124/dmd.32.5.512.

[23] M Hennessy, D Kelleher, JP Lloyd, A Alrajhi, O Meenaghan, C McDonald, F Mulcahy, JP Spiers, and J Feely. St John’s wort increases expression of P-glycoprotein: implications for drug interactions. *British Journal of Clinical Pharmacology*, 53(1):75–82, 2002. doi: 10.1046/j.1365-2125.2002.01512.x.

[24] H Assefa and V Butterweck. The role of hyperforin in the metabolic and transport-mediated drug interactions of St. John’s wort. *Planta Medica*, 70(4):291–300, 2004. doi: 10.1055/s-2004-818938.

[25] Er-Jia Wang, Mary Barecki-Roach, and William W Johnson. Quantitative characterization of direct P-glycoprotein inhibition by St John’s wort constituents hypericin and hyperforin. *Journal of Pharmacy and Pharmacology*, 56(11):1451–1456, 2004. doi: 10.1211/0022357044736.

[26] C Quiney, C Billard, A M Faussat, C Salanoubat, and J P Kolb. Hyperforin directly inhibits AKT1 kinase activity and promotes apoptosis in AML cells. *Leukemia*, 21(10):2101–2111, 2007. doi: 10.1038/sj.leu.2404834.

[27] C Quiney, C Billard, C Salanoubat, J D Fourneron, and J P Kolb. Hyperforin inhibits MMP-9 secretion by B-cell chronic lymphocytic leukemia cells. *Leukemia*, 20(8):1514–1521, 2006. doi: 10.1038/sj.leu.2404283.

[28] David Mendez, Anna Gaulton, A Patrícia Bento, Jon Chambers, Marleen De Veij, Eloy Félix, María Paula Magaña, Juan F Mosquera, Prudence Mutowo, Michał Nowotka, Maria Gordillo-Marañón, Fiona Hunter, Laura Junco, Grace Mugumbate, Milagros Rodriguez-Lopez, Francis Atkinson, Nicolas Bosc, Chris J Radoux, Aldo Segura-Cabrera, Anne Hersey, and Andrew R Leach. ChEMBL: towards direct deposition of bioassay data. *Nucleic Acids Research*, 47(D1):D930–D940, 2019. doi: 10.1093/nar/gky1075.

[29] UniProt Consortium. UniProt: the Universal Protein Knowledgebase in 2023. *Nucleic Acids Research*, 51(D1):D483–D489, 2023. doi: 10.1093/nar/gkac1052.

- [30] Kristian Leuner, Viacheslav Kazanski, Marina Müller, Kirill Essin, Britta Henke, Martina Gassen, Christopher Koch, Christina Bulut, Karola Silbermann, Annette Kopp-Schneider, Gerald Thiel, Vladimir Laketa, Inna Gorshkova, Valentina Przetchskikh, Christian Harteneck, Wolfgang F Graier, Vadym Degtiar, Peter Lipp, Axel Lückhoff, and Walter E Müller. Hyperforin—a key constituent of St. John’s wort specifically activates TRPC6 channels. *The FASEB Journal*, 21(14):4101–4111, 2007. doi: 10.1096/fj.07-8110com.
- [31] Katarina Hostanska, J Reiher, S Jessenmeyer, J Reichling, and R Saller. Hyperforin and hypericin: synergistic cytotoxicity and induced apoptosis in human malignant cell lines. *European Journal of Pharmaceutics and Biopharmaceutics*, 55(3):301–310, 2003. doi: 10.1016/s0939-6411(03)00021-3.
- [32] Vikas Kumar, Alexander Mdzinarishvili, Thomas Kiewert, Maria P Abbracchio, Annalisa Pinna, Renata Ciccarelli, Walter E Müller, and Jochen Klein. NMDA receptor-antagonistic properties of hyperforin, a constituent of St. John’s wort. *Journal of Pharmacological Sciences*, 102(1):47–54, 2006. doi: 10.1254/jphs.fp06041.
- [33] Belinda Phipson and Gordon K Smyth. Permutation P-values should never be zero: calculating exact P-values when permutations are randomly drawn. *Statistical Applications in Genetics and Molecular Biology*, 9(1):Article 39, 2010. doi: 10.2202/1544-6115.1585.
- [34] Justin K Huang, Daniel E Carlin, Michael Ku Yu, Wei Zhang, Jason F Kreisberg, Pablo Tamayo, and Trey Ideker. A systematic evaluation of molecular networks for discovery of disease genes. *Cell Systems*, 6(4):484–495, 2018. doi: 10.1016/j.cels.2018.03.001.
- [35] RDKit. RDKit: Open-source cheminformatics. <https://www.rdkit.org>, 2026. Version 2026.03.3.

Declarations

Funding

This research received no external research funding.

Ethics declarations

This study uses only publicly available databases and does not involve human participants, human tissue, or animal experiments; no ethics approval was required.

Acknowledgements

The authors acknowledge help from AI assistant tools for development and analysis, as disclosed in the Methods section.

Author contributions

A.B. conceived and designed the study, developed the analytical framework, curated the data, conducted the computational and statistical analyses, generated the figures, interpreted the results, and drafted the initial manuscript. S.M. contributed to data verification, methodological discussion, refinement of result presentation, and critical revision of the manuscript. V.T.M. supervised the study, provided scientific and methodological guidance, contributed to interpretation of the findings, and critically revised the manuscript. All authors reviewed and approved the final manuscript.

Competing interests

The authors declare no competing interests.

Additional information

Correspondence and requests for materials should be addressed to V.T.M.

Email: vijayakm2@srmist.edu.in.

ORCID iDs: Antony Bevan, 0009-0002-3676-2042; Vijayakumar Thangavel Mahalingam, 0000-0002-7749-3224.

Figures

Network context: target count and proximity to DILI genes

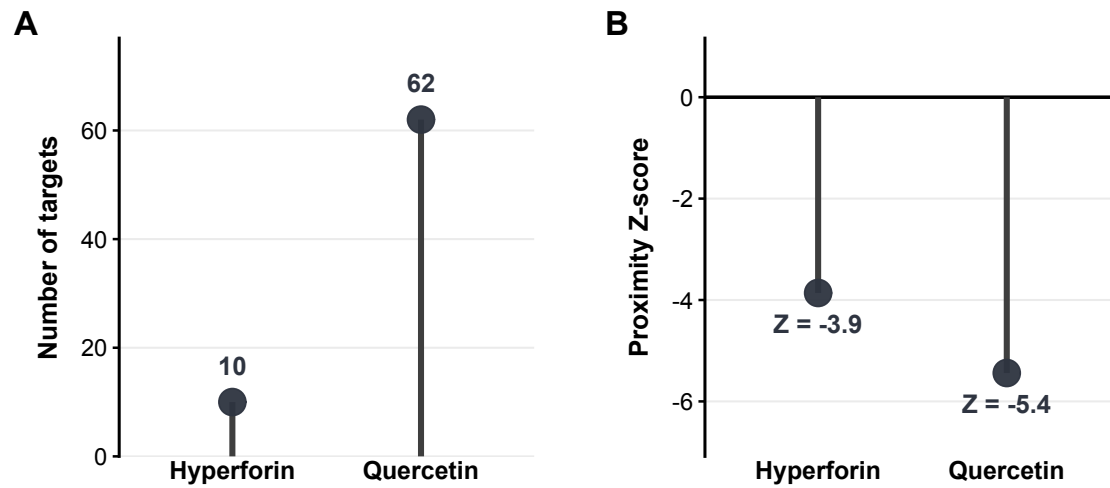


Figure 1. Network context. Target count and shortest-path proximity (d_c) of Hyperforin (10 targets) and Quercetin (62 targets) to the 82-gene DILI module. Proximity Z-scores are deviations from a degree-matched null ($n = 1,000$); negative values indicate closer-than-random proximity. STRING v12.0 (≥ 900), GTEx v8 liver LCC.

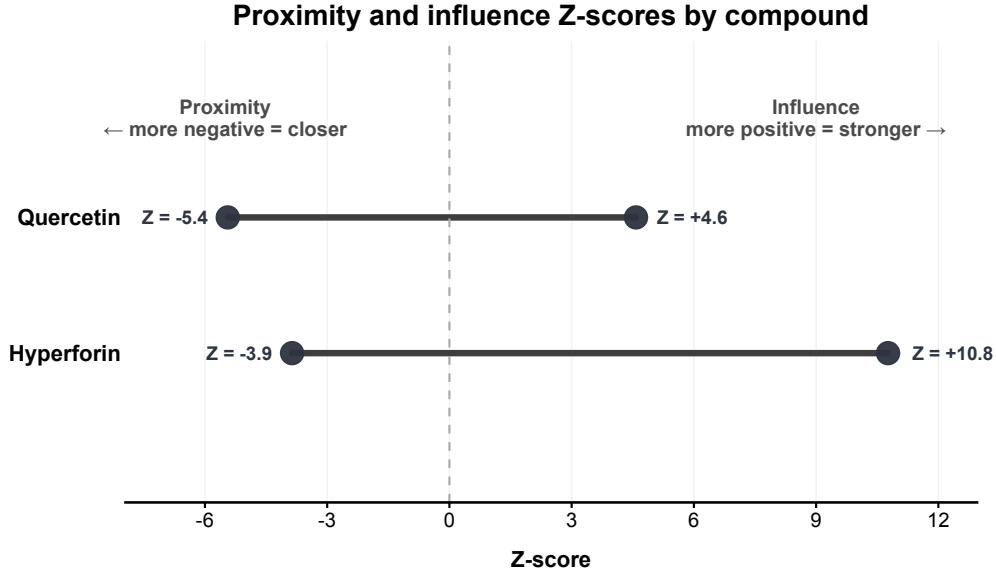


Figure 2. Effect size and evidence dissociate. At $\text{STRING} \geq 900$, Hyperforin is closer to the DILI module ($d_c = 1.30$ vs 1.68) yet has the weaker proximity Z-score (-3.86 vs -5.44); per-target influence ranks Hyperforin higher. Lines connect each compound’s proximity and influence statistics. Degree-matched permutation nulls ($n = 1,000$).

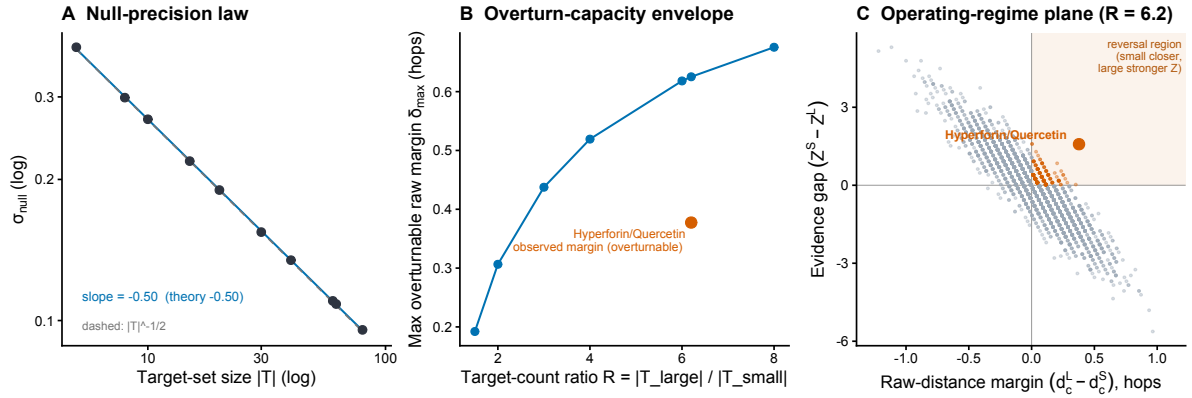


Figure 3. The null-precision law defines a conditional operating regime for rank reversal. Degree-controlled target-set probes on the liver LCC and DILI module (Methods); no toxicity outcome is modelled. **(A)** The null standard deviation scales as $\sigma_{\text{null}} \propto |T|^{-1/2}$ on the real DILI module (fitted slope -0.50 ; dashed $-1/2$ reference); the same exponent holds for random size-matched pseudo-modules (Table S4). **(B)** The maximum raw-distance margin the larger set can overturn, $\delta_{\text{max}}(R)$, increases with target-count ratio but saturates for fixed smaller-set size; the Hyperforin/Quercetin observed margin (0.38 at $R = 6.2$) lies below the envelope and is therefore overturnable. **(C)** Operating-regime plane at $R = 6.2$: each point is a probe pair, plotted by raw-distance margin (> 0 : smaller set closer) versus evidence gap (> 0 : larger set stronger); the reversal region is shaded. Material-margin reversals are rare for generic probes; Hyperforin/Quercetin (orange) sits in the reversal region at the 91st proximity percentile.

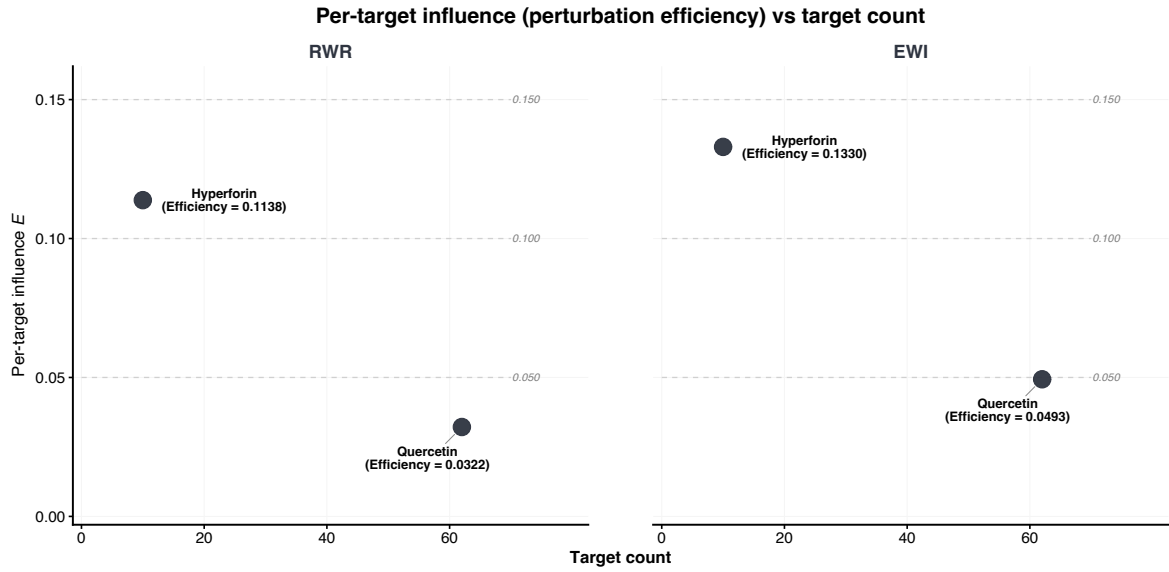


Figure 4. Perturbation efficiency. Per-target influence (perturbation efficiency E , Eq. (3)) versus target count; horizontal lines mark constant-influence tiers. E is the mean per-target influence on the DILI module and is comparable across compounds of different target count.

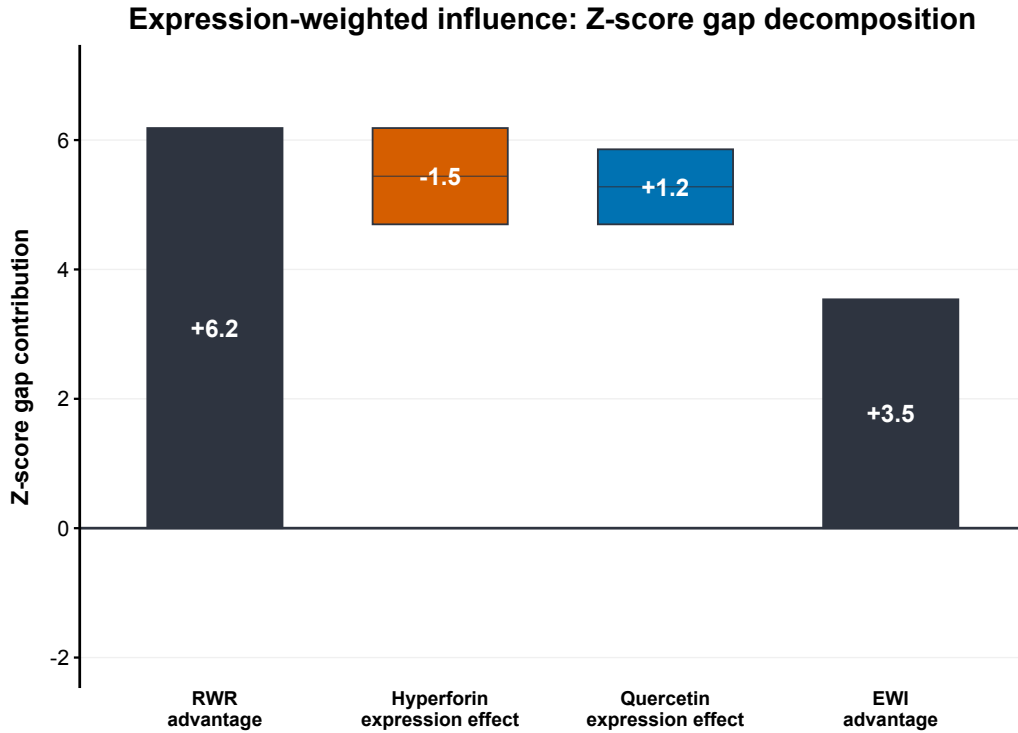


Figure 5. Expression-weighted influence. Waterfall decomposition of the RWR Z-score gap (Hyperforin – Quercetin) under destination-node liver-expression weighting: the gap narrows from +6.19 (RWR) to +3.54 (EWI): Hyperforin attenuates (–1.49) through liver-active hubs while Quercetin gains (+1.16) from high-expression nodes, but the ranking persists and is insensitive to the expression floor. GTEx v8 liver, STRING ≥ 900 , $n = 1,000$.

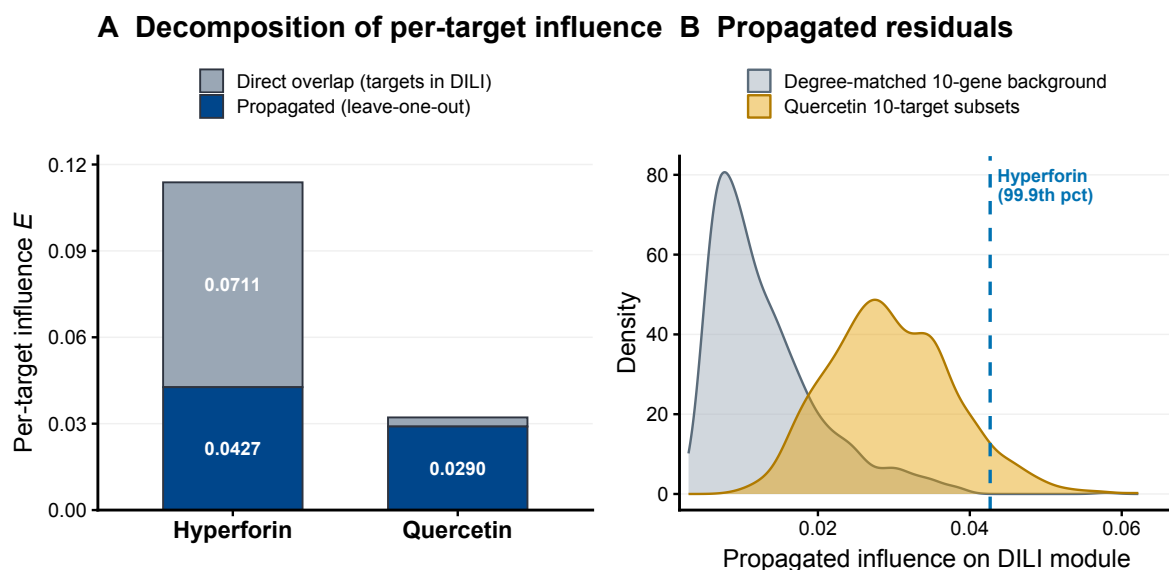


Figure 6. Direct versus propagated decomposition of per-target influence (STRING ≥ 900). (A) Per-target influence E separates into a direct-overlap component (steady-state mass on target $\cap$ DILI nodes) and a propagated component isolated by leave-one-out; 62% of Hyperforin’s influence is direct overlap, and the two compounds’ propagated components are close (0.0427 vs 0.0290). (B) The propagated component against null models: Hyperforin (dashed line, 99.9th percentile) exceeds a degree-matched random 10-gene background but lies within the spread of size-matched Quercetin 10-target subsets ($n = 1,000$ each), i.e. above background yet not unique.

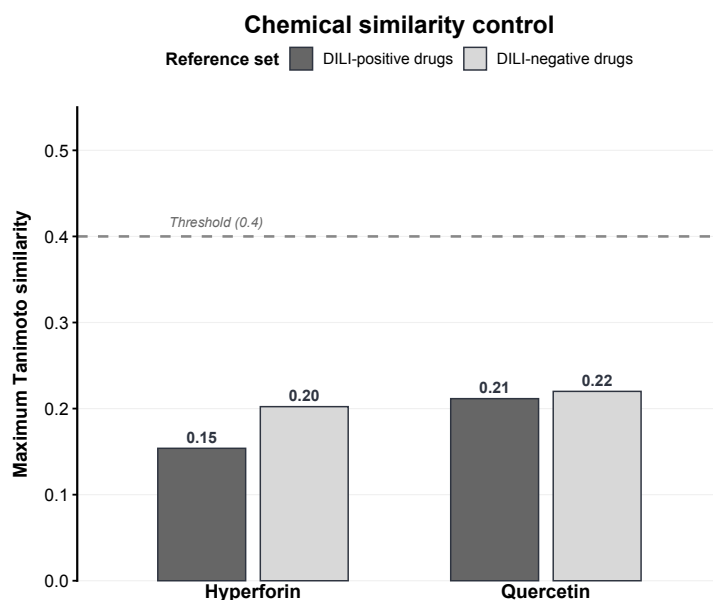


Figure 7. Chemical-similarity control. Maximum Tanimoto similarity to DILIrank 2.0 reference drugs (retrievable-SMILES binary subset: 542 DILI-positive [vMost/vLess-concern] and 365 DILI-negative [vNo-concern]; 354 ambiguous-concern drugs excluded); both compounds fall below the 0.4 structural-analogue threshold, arguing against close structural confounding within the DILIrank reference set.

Supplementary Information

Separating effect size from statistical evidence in network-proximity rankings under target-count asymmetry

Antony Bevan, Sarputheen M, and Vijayakumar Thangavel Mahalingam

Supplementary Notes

Network type. STRING combined-confidence edges integrate multiple evidence channels (experiments, curated databases, co-expression, genomic context, text mining); the resulting graph is a *functional association network*, not a purely physical protein-protein binding network. We use “interactome” in this broad STRING association-network sense throughout.

Text-mining robustness. Because the DILI module is DisGeNET-derived (literature-based) and the STRING combined score includes a text-mining channel, some target-DILI edges could reflect literature co-mention rather than independent evidence. Following Huang et al. [34], we rebuilt the interactome after removing every edge supported *solely* by the STRING text-mining channel (all other evidence channels zero) and re-ran all three metrics; a from-scratch rebuild without this filter recovers all committed network edges (0 missing, with a small mapping superset), and the full-rebuild observed values match the committed values, so the comparison is mapping-consistent. Text-mining-only edges constitute 0.7% (≥ 900) to 2.4% (≥ 700) of the liver-expressed LCC. Removing them leaves the effect ordering intact: Hyperforin remains closer ($d_c = 1.30$ vs 1.67 at ≥ 900) and higher in per-target random-walk (0.114 vs 0.033) and expression-weighted (0.134 vs 0.050) influence, with the ordering preserved on all three metrics at both thresholds (all $p < 0.002$; per-metric values in Table S5, full output in `results/tables/string_textmining_sensitivity.csv`). This argues against literature text-mining circularity as the driver of the effect/evidence dissociation, consistent with Huang et al.’s finding that STRING’s disease-gene signal is robust to text-mining removal.

DILI-module provenance. The DILI gene set is derived from DisGeNET curated gene-disease associations (UMLS C0860207). It is a *curated association module*, not a validated causal gene set: some members are metabolism genes, pharmacogenes, or literature-associated biomarkers rather than established causal DILI drivers. This is relevant because four Hyperforin targets (NR1H2, CYP2C9, ABCB1, MMP2) are themselves members of this module; the direct-propagated decomposition (Table S9) quantifies the resulting direct overlap.

Null estimands. Two degree-matched permutation nulls are reported. The *fixed-disease null* randomises only the target set and holds the disease module at the 82 real DILI genes; it answers “are these targets unusually close to *this* disease module?” and is the appropriate primary null for comparing two compounds against the same disease. The *two-sided null* (after Guney et al. [4]) additionally randomises a degree-matched disease set; it answers the population-level drug-disease question and is reported as a fidelity sensitivity (Table S13).

p-value convention. All permutation p-values are empirical, directional, and computed as $(r + 1)/(n + 1)$ [33]: one-sided for *smaller-than-null distance* (proximity) and *larger-than-*

null influence (RWR/EWI). With $n = 1,000$ the floor is 9.99×10^{-4} . Z-scores are the primary magnitude report; no two-tailed permutation p-value is used.

Supplementary Figures

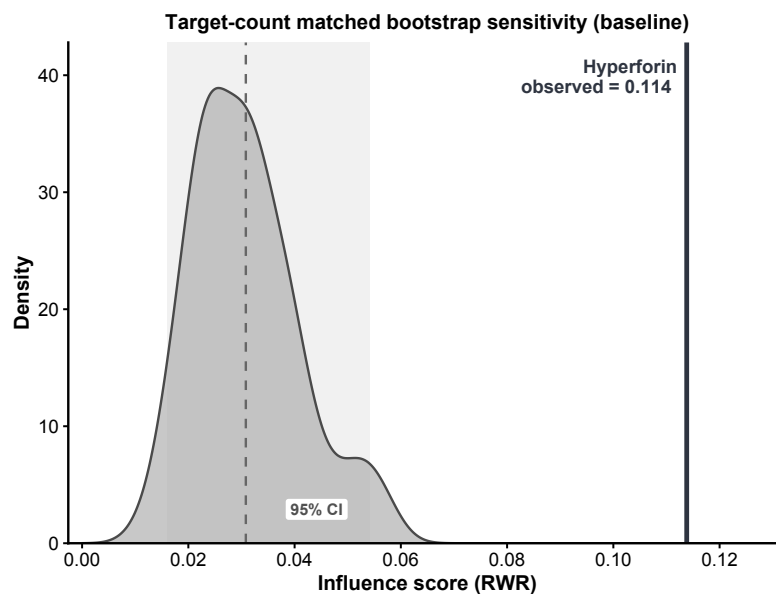


Figure S1. Bootstrap sensitivity (baseline). Distribution of RWR influence for random 10-target subsets of Quercetin; solid line, Hyperforin observed influence. This baseline control does not adjust for target–DILI overlap; the leakage-controlled analysis (Table 3, Fig. 6) supersedes the headline ratio. STRING ≥ 900 .

Supplementary Tables

All values above are regenerated by the committed pipeline and reviewer-evidence scripts; ratios, standardized scores, and separation values are computed from full-precision underlying data and may differ in the last displayed digit from the quotient of the rounded entries shown; cross-document numerical consistency is enforced by `verify_numbers.py`.

Table S1. Target curation evidence. Hyperforin targets are literature-curated and span a direct nuclear receptor (PXR/NR1I2), PXR-induced downstream enzymes, transporters, and direct-inhibition targets; Quercetin targets are uniformly ChEMBL bioactivity-derived. This documents the evidentiary asymmetry whose downstream consequence is audited in Table S9.

Gene	Source	Evidence type	Direct	In DILI	Shared	Kept [†]
<i>Hyperforin (literature-curated)</i>						
NR1I2	literature	direct receptor agonist (PXR)	yes	yes	no	yes
CYP3A4	literature	PXR-induced enzyme	no	no	yes	no
CYP2C9	literature	enzyme (inhib./induction)	partial	yes	no	no
CYP2B6	literature	PXR-induced enzyme	no	no	no	no
ABCB1	literature	transporter (induced + inhib.)	yes	yes	no	no
ABCG2	literature	transporter modulation	partial	no	yes	no
ABCC2	literature	transporter modulation	partial	no	no	no
AKT1	literature	direct kinase inhibition	yes	no	yes	yes
MMP2	literature	secretion inhibition	partial	yes	yes	yes
MMP9	literature	secretion inhibition	partial	no	yes	yes
<i>Quercetin (ChEMBL bioactivity, ChEMBL159; 62 targets, full list Table S14)</i>						
in vitro binding/activity ($IC_{50}/K_i/EC_{50} \leq 10 \mu M$)			yes	MMP2 (1)	5	n/a

“Direct” classifies the target–compound interaction: *yes* = direct molecular target (receptor agonism or enzyme/kinase inhibition); *partial* = mixed or context-dependent direct engagement (e.g. both induction and inhibition); *no* = downstream induced effect rather than direct binding. [†]Kept = retained in the “CYP/transporter-out” leakage control (Table S9).

Table S2. Hyperforin targets in the liver-expressed LCC (STRING ≥ 900). Node degrees are recomputed in the analysis network (correcting earlier values taken from a larger network). TPM from GTEx v8 liver.

Gene	Protein	TPM	Degree	DILI link
NR1I2	PXR	43	4	direct (module member)
CYP3A4	CYP3A4	335	73	indirect
CYP2C9	CYP2C9	434	52	direct (module member)
CYP2B6	CYP2B6	125	34	indirect
AKT1	PKB	33	128	indirect
ABCB1	P-gp	7	1	direct (module member)
ABCC2	MRP2	60	1	indirect
ABCG2	BCRP	4	2	indirect
MMP2	MMP-2	5	20	direct (module member)
MMP9	MMP-9	1	25	indirect

Table S3. Permutation null parameters across metrics (STRING ≥ 900 ; $n = 1,000$ degree-matched permutations). The null standard deviation σ_{null} of the 62-target compound is $\approx 2.5\times$ smaller than that of the 10-target compound for shortest-path, random-walk *and* expression-weighted influence alike — the Law-of-Large-Numbers shrinkage ($\sqrt{62/10} = 2.49$), identical across metrics. This is why Z-score magnitude is not comparable across compounds of very different target count.

Metric	Compound	μ_{null}	σ_{null}	x_{obs}	Z
Shortest-path d_c	Hyperforin (10)	2.2086	0.2353	1.3000	−3.86
	Quercetin (62)	2.1749	0.0914	1.6774	−5.44
RWR influence	Hyperforin (10)	0.0141	0.0093	0.1138	10.77
	Quercetin (62)	0.0148	0.0038	0.0322	4.58
EWI influence	Hyperforin (10)	0.0196	0.0122	0.1330	9.28
	Quercetin (62)	0.0210	0.0049	0.0493	5.74

σ_{null} ratio (Hyperforin/Quercetin): shortest-path 2.57, RWR 2.45, EWI 2.47 (expected $\sqrt{62/10} = 2.49$).

Table S4. Operating-regime benchmark: margin-conditional rank-reversal rates (liver LCC ≥ 900 , DILI module; probes pinned to the full compound-target row degree profile, with DILI-module genes excluded from the candidate pool; smaller set $|T_S| = 10$; 500,000 probe pairs per row, seed 42). Each entry is the fraction of pairs in which the smaller set is genuinely closer by a material raw margin ($\geq \delta_0$ hops) yet the larger set has the more negative Z-score. Reversals are rare and require a smaller set above the 90th percentile of probe-pair margins: a no-shrinkage counterfactual ($\sigma_L := \sigma_S$) gives 0% at every R , and unconditionally the Hyperforin/Quercetin directional pattern (smaller set closer *and* larger set stronger Z) occurs in only 6.5% of exact $R = 6.2$ pairs, collapsing to near-zero once a material margin is required. The null-SD exponent is -0.499 (95% CI $[-0.502, -0.495]$, $R^2 = 0.9999$) and similar across the DILI module and three size-matched pseudo-modules (DILI -0.495 ; three random size-matched pseudo-modules mean -0.498).

$R = T_L / T_S $	(T_S , T_L)	reversal, $\delta_0 \geq 0.3$ (%)	reversal, $\delta_0 \geq 0.5$ (%)
1.5	(10, 15)	0.00	0.00
2.0	(10, 20)	0.00	0.00
3.0	(10, 30)	0.00	0.00
4.0	(10, 40)	0.00	0.00
6.0	(10, 60)	0.10	0.00
6.2	(10, 62)	0.06	0.00
8.0	(10, 80)	0.39	0.00

Table S5. Text-mining robustness: metric values before and after removing text-mining-only STRING edges (liver-expressed LCC; following Huang et al. [34], every edge supported *solely* by the STRING text-mining channel was removed). “Committed” is the value in the main analysis; “no-TM” is after removal. The Hyperforin > Quercetin ordering (smaller d_c ; larger influence) is preserved on all three metrics at both network thresholds, and every no-TM permutation $p < 0.002$; arguing against literature text-mining circularity as the driver. Source: results/tables/string_textmining_sensitivity.csv.

Metric	STRING	Hyperforin		Quercetin		winner
		committed	no-TM	committed	no-TM	
d_c (smaller closer)	≥ 700	0.6000	0.6000	1.3387	1.3387	Hyp
	≥ 900	1.3000	1.3000	1.6774	1.6721	Hyp
RWR influence (larger)	≥ 700	0.1212	0.1243	0.0326	0.0335	Hyp
	≥ 900	0.1138	0.1144	0.0322	0.0329	Hyp
EWI influence (larger)	≥ 700	0.1449	0.1472	0.0511	0.0516	Hyp
	≥ 900	0.1330	0.1337	0.0493	0.0501	Hyp

Table S6. Threshold robustness (STRING ≥ 700 vs ≥ 900 liver LCC; $n = 1,000$, seed 42). The proximity- Z ranking reverses across thresholds — Hyperforin is the more “significant” at ≥ 700 , Quercetin at ≥ 900 — whereas the raw closest distance (d_c , Hyperforin closer) and the per-target influence (E , Hyperforin higher) rankings are stable at both thresholds. This is the effect-size/evidence dissociation of the main text, shown across network construction.

Metric	Compound	STRING ≥ 700		STRING ≥ 900	
		value	Z	value	Z
d_c (closest distance)	Hyperforin	0.60	−6.04	1.30	−3.86
	Quercetin	1.34	−5.46	1.68	−5.44
E (RWR influence)	Hyperforin	0.1212	+11.04	0.1138	+10.77
	Quercetin	0.0326	+5.46	0.0322	+4.58
E (EWI influence)	Hyperforin	0.1449	+10.35	0.1330	+9.28
	Quercetin	0.0511	+7.01	0.0493	+5.74

Table S7. Restart-probability (α) sensitivity (STRING ≥ 900). The Hyperforin > Quercetin per-target-efficiency ranking holds at every α ; the ratio is α -dependent and is not interpreted as an absolute constant.

α	0.10	0.15	0.20	0.30	0.50	0.70
E Hyperforin	0.0907	0.1138	0.1344	0.1716	0.2390	0.3032
E Quercetin	0.0312	0.0322	0.0322	0.0310	0.0271	0.0227
ratio	2.90	3.54	4.18	5.53	8.81	13.35

Table S8. Expression-floor sensitivity (EWI) (STRING ≥ 900). The 0.01 floor is immaterial; the ratio moves only in the third decimal.

floor	0	10^{-3}	10^{-2}	5×10^{-2}	10^{-1}
E Hyperforin	0.1330	0.1330	0.1330	0.1330	0.1327
E Quercetin	0.0493	0.0493	0.0493	0.0493	0.0491
ratio	2.695	2.695	2.695	2.695	2.702

Table S9. Direct-propagated decomposition of per-target influence: full controls (STRING ≥ 900). Four of ten Hyperforin targets are DILI-module members vs one of 62 Quercetin targets; isolating the propagated component (leave-one-out) and removing each source of direct overlap attenuates the per-target advantage to a modest residual.

Control	$E(\text{Hyp})$	$E(\text{Quer})$	ratio
Baseline (includes overlap)	0.1138	0.0322	3.54
Self-mass excluded	0.0427	0.0290	1.47
DILI-member seeds removed	0.0431	0.0295	1.46
CYP/transporter seeds removed	0.0338	0.0290	1.17
Shared targets removed (both)	0.0541	0.0288	1.88
<i>Propagated (leave-one-out) reference distributions ($n = 1,000$; Fig. 6B):</i>			
Global degree-matched random set ($n = 10$)	Hyp 0.0427 vs null mean 0.0130 (95th 0.0281; max 0.0580); 99.9th percentile, $3.3\times$ mean, $p = 0.002$, does not exceed all		
Size-matched Quercetin 10-subsets	Hyp 0.0427 vs subset mean 0.0296 (max 0.0622); $1.4\times$, overlapping		

Table S10. Network separation between target set and DILI module. Menche et al. separation measure $S_{AB} = \langle d_{AB} \rangle - (\langle d_{AA} \rangle + \langle d_{BB} \rangle) / 2$ [3] (STRING ≥ 900); $S_{AB} > 0$ indicates topological separation. Both target sets are separated from the DILI module, and Hyperforin is *more* separated despite its four direct-overlap targets, because its targets form a tighter cluster ($\langle d_{AA} \rangle = 1.20$ vs 1.55). This corroborates the decomposition (Table 3, Fig. 6): Hyperforin’s per-target influence advantage arises from a few targets that *are* DILI genes (direct overlap), not from its target set being embedded in the DILI neighbourhood.

Target set	$\langle d_{AB} \rangle$	$\langle d_{AA} \rangle$	$\langle d_{BB} \rangle$	S_{AB}
Hyperforin	1.699	1.200	1.500	0.349
Quercetin	1.644	1.548	1.500	0.119

Table S11. Direct (distance-1) DILI connectivity of Hyperforin targets (regenerated on the STRING ≥ 900 liver LCC). Connectivity is concentrated in the cytochrome-P450 axis; the transporters (ABCB1, ABCC2, ABCG2) and the receptor NR1I2 have no additional distance-1 DILI neighbours (NR1I2 and ABCB1 are themselves module members, at distance 0). Per-target means: **Hyperforin 3.40** direct DILI neighbours (34 connections across 10 targets) vs **Quercetin 1.53** (95 across 62) — a 2.2 $\times$ enrichment that complements the distance-0 direct-overlap decomposition (Table S9).

Hyperforin target	n	Direct DILI neighbours (distance 1)
CYP3A4	11	CYP2A6, CYP2C19, CYP2C9, CYP2E1, FMO3, GSTM1, GSTM2, GSTP1, NR1I2, NR1I3, UGT1A9
CYP2C9	9	CYP2A6, CYP2C19, CYP2E1, FMO3, GSTM1, GSTM2, GSTP1, PTGS2, UGT1A9
CYP2B6	6	CYP2A6, CYP2C19, CYP2C9, CYP2E1, PTGS2, UGT1A9
AKT1	4	CTNNB1, IGF1, MAP3K5, NFE2L2
MMP9	3	LCN2, MMP2, SPP1
MMP2	1	LCN2
ABCB1, ABCC2, ABCG2, NR1I2	0	—

Table S12. DILI-module sensitivity: removing the entangled pharmacogenes. Every other robustness check in this work varies the *method* (α , expression floor, network threshold, degree-binning, two nulls); this one varies the *disease module*, the locus of the target–disease circularity. We remove from the 82-gene module the four genes that are simultaneously Hyperforin targets and DILI members (ABCB1, CYP2C9, MMP2, NR1I2), giving a reduced module D' that the drug targets cannot directly overlap, and recompute the propagated (leave-one-out) per-target influence symmetrically for both compounds (STRING ≥ 900 , $\alpha = 0.15$). The propagated advantage does *not* collapse — it slightly increases, because Hyperforin’s leave-one-out value is unchanged (those genes were already excluded from its own scoring) while Quercetin loses propagation onto ABCB1/CYP2C9/NR1I2, which it had benefited from as non-target DILI genes. Against a degree-matched random 10-gene background scored on D' ($n = 1,000$), Hyperforin remains at the 99th percentile (3.45 $\times$ the background mean, empirical $p = 0.005$) but does not exceed every random set. The modest propagated signal supports the interpretation that the propagated residual is not eliminated after removing the overlapping pharmacogenes from the module.

Disease module	$ D $	Propagated E (Hyp)	Propagated E (Quer)	Ratio
Full DILI module	82	0.0427	0.0290	1.47
Overlap genes removed (D')	78	0.0427	0.0271	1.58

Table S13. Independent reimplementations of the canonical proximity toolbox: full null parameters (emreg00/toolbox, re-derived by GUNY_FIDELITY_check.py; STRING ≥ 900 ; $n = 1,000$; seed 42; $|D| = 82$ preserved). This is the detailed companion to the compact main-text summary (Table 4): for each null construction it reports the observed closest distance d_c , the null mean μ_{null} and standard deviation σ_{null} , and the resulting Z . The observed d_c reproduces the toolbox exactly (Hyperforin 1.300, Quercetin 1.677). The reimplementations’ $\pm 25\%$ window (group 1, Quercetin $Z = -5.35$) matches the committed **src** pipeline ($Z = -5.44$; Table S3) within permutation tolerance, confirming the two implementations agree. The $\pm 25\%$ window never silently falls back to unmatched nodes: every Hyperforin target retains ≥ 143 degree-matched candidates (its highest-degree target, AKT1, has degree 128). The evidence dissociation is clear under the fixed-disease null and attenuates to near-parity under the two-sided null; Hyperforin is the topologically closer compound throughout.

Null model	Compound	d_c	μ_{null}	σ_{null}	Z
$\pm 25\%$ window, fixed disease	Hyperforin	1.30	2.22	0.238	-3.86
	Quercetin	1.68	2.18	0.093	-5.35
Guney ≥ 100 -bin, fixed disease	Hyperforin	1.30	2.23	0.228	-4.09
	Quercetin	1.68	2.16	0.090	-5.34
Guney ≥ 100 -bin, two-sided	Hyperforin	1.30	2.05	0.212	-3.55
	Quercetin	1.68	2.01	0.090	-3.66

σ_{null} ratio (Hyperforin/Quercetin), from full-precision values: 2.56 ($\pm 25\%$ fixed), 2.53 (≥ 100 -bin fixed), 2.37 (≥ 100 -bin two-sided); expected $\sqrt{62/10} = 2.49$. The Law-of-Large-Numbers shrinkage therefore persists under every null construction, including Guney’s exact two-sided null.

Table S14. Quercetin target genes in the liver-expressed LCC (STRING ≥ 900 , GTEx v8 liver TPM ≥ 1 ; ChEMBL CHEMBL159). Sorted by descending TPM.

Gene	TPM	Gene	TPM	Gene	TPM	Gene	TPM
CFB	1115	CYP3A4	335	FN1	229	ALDH2	183
ANPEP	160	PPIA	112	SERPINA5	104	CYP1A2	72
CA2	64	APP	63	PYGL	55	HDAC6	45
ESRRA	42	MAOA	35	AKR1C2	33	AKT1	33
CTSH	28	XDH	26	CHRNA4	25	PIK3R1	24
PIM1	24	LDLR	23	ELOVL1	18	EGFR	17
PKN1	16	GSK3A	13	YES1	13	MET	12
DAPK1	12	BACE1	11	CSNK2A1	10	FSTL1	9
SIRT6	8	GSK3B	7	CDK7	7	CAV2	7
PTPN2	6	CYP1A1	5	PRMT7	5	MMP2	5
AKR1B1	5	PDE6D	5	BRAF	4	PTK2	4
ABCG2	4	IQGAP1	4	ADRB2	3	KDR	3
SRC	3	ALOX5	3	CYP1B1	3	TLR4	3
NUAK1	3	AXL	2	ADA	2	LCK	2
ABCC1	2	PLK1	1	ACHE	1	MMP9	1
SYK	1	PDZK1IP1	1				

Table S15. Curation provenance (stage counts, reproducible from the committed data). Targets are filtered to the intersection of the ≥ 700 and ≥ 900 liver LCCs, so target counts are identical at both thresholds; the DILI module is filtered per threshold.

Stage	Hyperforin	Quercetin	DILI module
Raw (lit. / ChEMBL / DisGeNET)	14	122	127
Processed (human, UniProt-mapped)	14	87	—
In ≥ 700 liver-expressed LCC	10	62	84
In ≥ 900 liver-expressed LCC	10	62	82

Table S16. DILI gene set (82 genes) in the liver-expressed STRING ≥ 900 LCC (DisGeNET, UMLS C0860207; GTEx v8 liver TPM ≥ 1), alphabetical. The four genes that are also Hyperforin targets (the distance-0 direct overlap, Table S9) are in bold.

82 DILI-associated genes							
ABCB1	AHR	ALB	ALDOB	AMBP	APOA1	APOE	APOH
ARG1	ARNT	ATG5	BAX	BTD	C3	CAT	CCL2
CLU	COL3A1	CTNNB1	CXCL1	CXCL10	CYP2A6	CYP2C19	CYP2C9
CYP2E1	DGAT2	ENO1	FGA	FLT1	FMO3	GADD45A	GC
GCLC	GPT	GSN	GSTM1	GSTM2	GSTP1	HLA-A	HLA-B
HLA-DQB1	HLA-DRB1	HMGB1	HMOX1	HPD	HPX	IGF1	IL18
IL1R2	KRT18	LCN2	LGALS3	MAP3K5	MED1	MMP2	MTHFR
NAT2	NFE2L2	NR1H3	NR1H4	NR1H2	NR1H3	PLAT	PLG
PNP	POLG	PON1	PPARA	PRKDC	PTGS2	RBP1	SLPI
SNX18	SOD1	SOD3	SPP1	TALDO1	TBXA2R	TCTN1	TF
TTR	UGT1A9						